

Journal Supplementary A — The accuracy-only view and why it fails

Companion to: *Beyond Accuracy: A Multi-Axis Evaluation Framework for Interpretable Topic Models on HSI*

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A.1 Scope

This supplement expands Section I of the journal paper. We provide (a) a longer historical account of how the HSI community arrived at the accuracy-only evaluation discipline, (b) five concrete examples in which a single OA number hides materially different model behaviours, (c) the philosophical commitment that distinguishes *interpretability* from *post-hoc explanation*, and (d) the framing that motivates a twelve-axis framework rather than a smaller or larger one.

B.2 How the field arrived at OA-as-summary

The earliest HSI classification papers [1,2] settled on overall accuracy (OA) and kappa as the canonical summary because the underlying task was assumed to be *closed-set supervised classification*: given a labelled benchmark with L classes, predict the class of each unlabelled pixel; report the fraction correct. Under this assumption OA is monotone in the quantity the practitioner cares about (predictive accuracy) and the kappa adjustment handles class imbalance. Subsequent work imported the OA discipline wholesale into *unsupervised* and *interpretability-motivated* settings where the quantity the practitioner cares about is different.

Linear unmixing [3,4] imported OA-style reporting via downstream classification of predicted abundance vectors. Deep classification papers [5,6] continued the OA tradition. Topic-model adaptations to HSI [7,8] inherited OA from both ancestors. The price of this inheritance is that the OA number cannot itself answer the question “*why is this topic basis interpretable?*” — because the question is not what OA measures.

C.3 Five concrete examples of OA hiding bad behaviour

Example 1: seed-collapsed ProdLDA. ProdLDA fits on Indian Pines under five seeds yield ARI standard deviation 0.003 across seeds (extremely stable) and per-seed ARI *below* 0.17 (weak class discrimination). A single ProdLDA OA number averaged over seeds would be honest but uninformative; the practitioner needs to see both the stability and the level.

Example 2: high-OA on weakly identified topics. KSC under a 13-class supervised baseline reaches OA > 0.9 on the canonical fold; the same KSC under our band-mask diagnostic returns paired ARI ≈ 0.01 . A KSC paper that reports only OA would *look* like a strong contribution; the topic basis driving the classification is, however, band-fragile.

Example 3: HIDSAG inversion. On the labelled scenes the topic-routed-soft classifier is statistically indistinguishable from the raw-spectrum logistic baseline. On HIDSAG the topic-logistic classifier sits at posterior $\mu = 0.316$ vs raw $\mu = 0.585$ with $P[\mu_{\text{topic}} > \mu_{\text{raw}}] = 0.0$. The labelled-scene OA number would lead a practitioner to deploy the topic representation on HIDSAG; the hierarchical model says that decision is wrong.

Example 4: coherence-vs-ARI flip. ProdLDA’s top-15-word coherence is the highest on Indian Pines ($c_v = 0.630$), but its cluster-vs-label ARI is the lowest (0.165). A practitioner who picks ProdLDA on the basis of coherence gets a different downstream behaviour than one who picks on the basis of clustering quality.

Example 5: SLIC vs topic-dominant cross-method disagreement. On Indian Pines SLIC-500 and the canonical topic dominant map have pairwise ARI of 0.039 — i.e. the two “unsupervised partitions” of the same scene disagree almost completely on which pixels belong to which segment. The OA of either method against ground truth does not surface this disagreement; the cross-method ARI matrix does.

D.4 Interpretability versus post-hoc explanation

There is a methodological distinction we want to preserve. *Interpretability* is a property of the model itself: a topic basis ϕ is interpretable if every element of the parameterisation has a meaning the practitioner can read off the parameter. *Post-hoc explanation* is a property of an attached diagnostic: a feature-attribution map or an inverse-classification view that approximates the model’s behaviour on a sample. Topic models on HSI are advertised as belonging to the first category, but in practice the published evaluations support only the second: a high-OA topic model with a band-fragile basis is providing post-hoc explanation, not interpretability.

The twelve-axis framework defended in the journal paper restores the first-category claim or denies it explicitly. A topic basis whose twelve-axis card has stable, coherence-meaningful, band-robust, and sample-stable rows admits the “ ϕ_k has a material-level meaning” reading. A topic basis whose card has any of those rows degenerate admits only the “ θ is a useful feature for a downstream classifier” reading — which is the post-hoc reading and is allowed, but should be advertised as such.

E.5 Why twelve axes, not five or fifty

The number twelve is not magic. It reflects the orthogonalisation of all the evaluation questions the present project has needed to ask over ~ 150 iterative cycles of working with the artefact set. We checked four alternative counts:

- **Five axes (the OA + four corollaries view).** Insufficient. Drops cross-method agreement, topic-label coupling, preprocessing robustness, cross-scene transfer.
- **Seven axes.** Drops preprocessing robustness (essential on HIDSAG) and cross-scene transfer (essential when sensor families overlap, e.g. AVIRIS Salinas / Salinas-A).
- **Twelve axes (selected).** The smallest set that yields a distinct numeric per axis on the present benchmarks and admits a deterministic builder per axis without overlapping builders.
- **Twenty-plus axes.** We considered splitting F-3 into inferential-stability + estimation-stability, F-5 into mask-set vs band-budget, F-12 into intra-paper vs inter-paper baselines. Doing so multiplies the artefact footprint without adding qualitatively new information.

The twelve-axis count is therefore the minimum that surfaces every distinct failure mode we have observed.

References

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